Chapter 23

# Communicable Diseases

1. Understanding Communicable Diseases
2. Common Communicable Diseases
3. Fighting Communicable Diseases
4. Emerging Diseases and Pandemics

## Lesson 1 — Understanding Communicable Diseases

A **communicable disease** spreads from one living organism to another or through the environment. It happens when **pathogens** enter your body and multiply, causing an **infection**.

| | |
|---|---|
| **Viruses** | Invade cells and hijack them to replicate. Antibiotics DO NOT kill viruses. Examples: cold, flu, COVID-19, HIV |
| **Bacteria** | Single-celled; some produce toxins. Many treatable with antibiotics — but resistance is a growing crisis. |
| **Fungi** | Cause infections like athlete's foot and ringworm. Treated with antifungals. |
| **Protozoa** | Single-celled organisms causing diseases like malaria — transmitted by mosquitoes; kills hundreds of thousands globally. |

## Lesson 2 — Common Communicable Diseases

| | |
|---|---|
| **Common Cold** | 200+ different viruses; spreads via droplets and contaminated surfaces; no antibiotic cure |
| **Influenza (Flu)** | Viral; hits harder and faster than a cold; antivirals like Tamiflu reduce severity if taken within 48 hours |
| **Strep Throat** | Bacterial; treatable with antibiotics; untreated can cause rheumatic fever |

| | |
|---|---|
| **Hepatitis C** | NOW CURABLE with direct-acting antivirals (DAAs) — >95% cure rate in 8–12 weeks. Major update from original textbook. |

## Lesson 3 — Fighting Communicable Diseases

Vaccines train your immune system to recognize and fight specific pathogens without causing the disease. They have prevented more death and disability than almost any other medical intervention in history.

| | |
|---|---|
| **Live-attenuated** | Uses weakened pathogen; strong, often lifelong immunity. Examples: MMR, chickenpox |
| **Inactivated** | Uses dead pathogen. Examples: flu shot, hepatitis A |
| **Toxoid** | Uses inactivated toxin. Examples: tetanus, diphtheria |
| **mRNA (NEW since 2011)** | Genetic instructions teach cells to make a harmless piece of pathogen. Examples: COVID-19 vaccines (Pfizer, Moderna) |

## Lesson 4 — Emerging Diseases and Pandemics

**COVID-19** — the biggest pandemic since 1918 — killed over 7 million people worldwide. The mRNA vaccines (developed in ~11 months using decades of prior research) were authorized in December 2020. **Mpox** caused a global outbreak in 2022 and a renewed WHO emergency in August 2024.

# Chapter Vocabulary

## Lesson 1

| | |
|---|---|
| **Communicable disease** | A disease that spreads from one organism to another or through the environment |
| **Infection** | The condition that results when pathogens multiply in the body and damage cells |
| **Vector** | An organism (such as a mosquito or tick) that carries and transmits pathogens |

## Lesson 2

| | |
|---|---|
| **Pneumonia** | An infection of the lungs in which air sacs fill with fluid; can be viral or bacterial |
| **Jaundice** | Yellowing of the skin and eyes; a common symptom of hepatitis |

## Lesson 3

| | |
|---|---|
| **Immune system** | The body's network of cells, tissues, organs, and chemicals that fight pathogens |
| **Inflammatory response** | The immune system's first response to tissue damage or infection |
| **Antibody** | A protein produced by B cells that targets a specific antigen, marking it for destruction |
| **Vaccine** | A preparation that stimulates the immune system to develop protection against a disease |
| **Immunity** | The state of being protected against a specific disease |

## Lesson 4

| | |
|---|---|
| **Emerging infections** | Communicable diseases whose occurrence in humans has increased significantly in recent decades |
| **Pandemic** | A global outbreak of an infectious disease |
| **Epidemic** | A disease outbreak that affects many people in the same place at the same time |